Chapter 3

Technology and Engineering Teaching and Learning Process

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3.1. Introduction

Teaching and learning feed each other; teachers learn from learners shaping future lessons based on questions or struggles. Learners teach themselves and others by explaining ideas, solving problems, or collaborating in teams. If you are an active participant in your learning process since first grade or before, most likely you have not had any formal training in it, you learned what your parents or teacher thought you by repeating. Becoming an expert learner requires not only that you devote time and energy to learning, but that you devote time and energy into

learning how to learn. According to expertise, there are three processes of acquiring knowledge, cognitive learning, motor action learning, and affective learning. Here are the differences in which will be discussed. The teaching and learning process is a dynamic cycle where knowledge is transferred, discovered, challenged, and internalized. At its best, it's not just about passing information; it's about sparking curiosity, building skills, and transforming understanding into action.

3.2. Core Components of Teaching

- Instructional design content delivery assessment and feedback adaptability:
    - Planning goals, materials, and activities that suit learners' needs.
    - Explaining concepts through lectures, demonstrations, or storytelling
    - Evaluating understanding and guiding improvement through questions, tests, or projects.
    - Adjusting methods based on how learners respond or struggle.

- Assessment and feedback adaptability:
    - Evaluating understanding and guiding improvement through questions, tests, or projects.
    - Adjusting methods based on how learners respond or struggle.

- Essentials of Learning
    - Engagement: Staying mentally active and curious about the topic
    - Practice: Reinforcing knowledge through application and iteration.
    - Reflection: Thinking critically about what you've learned and how it applies
    - Transfer: Applying knowledge to new problems, contexts, or fields

The Three Domains of Learning

Bloom's Taxonomy divides learning into three major domains:

- **Cognitive**: Mental skills and knowledge.
- **Affective:** Emotions, attitudes, and values
- **Psychomotor:** Physical skills and coordination

Cognitive Learning

Cognitive learning is established by information recall and a higher level of scholarly skills. Since acquiring new knowledge and intellectual skills will be the primary focus of your engineering education, it's valuable to explore the process in some detail.

The research Bloom's Taxonomy of Educational objectives mention 6 levels of intellectual skills inside cognitive learning. **Bloom's Taxonomy** is a powerful framework for understanding how people learn and how educators can guide that learning. Developed by Benjamin Bloom and colleagues in 1956, it categorizes educational objectives into levels of complexity and specificity, helping teachers design lessons, assessments, and learning outcomes that build deeper understanding over time.

The most widely used part of Bloom's Taxonomy is the **cognitive domain**, which was revised to emphasize active learning. It includes six hierarchical levels:

- **Understand:** Explain ideas and concepts.
    - summarize, interpret, classify.
- **Apply:** Use information in new situations to implement, solve, and demonstrate.
    - implement, solve, demonstrate
- **Analyze:** Break down information into parts
    - compare, differentiate, examine
- **Evaluate:** Justify decisions or opinions.
    - critique, assess, defend
- **Create:** Produce new or original work.
    - design, invent, compose

Each level builds the previous one, before you can analyze, you must understand; before you can create, you must evaluate.

Cognitive in Engineering Technology

For someone like you, exploring engineering technology, Bloom's Taxonomy can guide how you:

- **Master fundamentals** (Remember & Understand)
- **Apply tools and techniques** (Apply)
- **Troubleshoot systems** (Analyze)
- **Evaluate design choices** (Evaluate)
- **Innovative solutions** (Create)

Back in high school you mainly worked at the first two levels of demonstrating subject mastery remembering and understanding. Often you could earn good grades by memorizing material and successfully repeating it back on exams or other assignments. In your college education, the expectations are higher. From the start, you will be expected to think at levels 3 and 4, applying and analyzing, mastering concepts and applying them to solve new problems. Your performance will be evaluated on your ability to apply what you learn in one context to new contexts. In your first two years you may be given some problems that require you to think at levels 5 and 6, evaluating and creating, but the strongest focus on these levels of thinking will come in your junior and senior years. Mastering the learning skills described above will help you become an excellent learner in future.

3.3. Affective Learning

Affective learning is all about how emotions, values, and attitudes shape the way we learn. It's one of the three domains in Bloom's Taxonomy, alongside cognitive (thinking) and psychomotor (doing) and it focuses on the **emotional dimension** of education. This is demonstrated by behaviors indicating attitudes of awareness, interest, attention, concern, and responsibility. This domain relates to emotions,

attitudes, appreciations, and values, such as enjoying, respecting, and supporting. The ability to listen and respond in interactions with others is part of affective learning.

Key Characteristics of Affective Learning

- Involves **feelings, emotions, and attitudes**
- Encourages **empathy, motivation, and engagement**
- Supports **value formation** and ethical reasoning
- Enhance **interpersonal skills** and emotional intelligence

The Five Levels of the Affective Domain (Krathwohl's Taxonomy)

- **Receiving:** Being aware and willing to listen or experience.
  - Example behaviors: Attending a lecture, watching a film.
- **Responding:** Actively participating or reacting to stimuli.
  - Example behaviors: Asking questions, joining discussions.
- **Valuing:** Seeing worth in something and expressing it.
  - Example behaviors: Supporting a cause, showing respect.
- **Organizing:** Integrating new values into one's belief system
  - Example behaviors: Prioritizing study over leisure
- **Characterizing:** Acting consistently with internalized values.
  - Example behaviors: Leading with integrity, showing empathy

Each level reflects deeper emotional engagement and internalization of values.

Affective In Engineering Technology

For someone like you, affective learning can help you:

- Stay **motivated** through tough problem-solving
- Develop **ethical reasoning** in design decisions
- Collaborate with empathy in **team-based projects**
- Reflect on the **social impact** of engineering solutions

Would you like help designing a learning activity or project that integrates affective learning into your engineering studies? I can help you build something meaningful and motivating.

3.4. Psychomotor learning

Psychomotor learning is all about the **connection between mind and movement**, where cognitive understanding meets physical action. Psychomotor learning is demonstrated by physical skills, synchronization, dexterity, manipulation, grace, strength, and speed, for example, actions that demonstrate the fine motor skills such as use of precision instruments or tools, or actions that exhibit motor skills such as dance or athletic performance.

What does psychomotor learning mean?

Psychomotor learning refers to the development of physical skills that require coordination, precision, and control. It's how we learn to do things like:

- Soldering a circuit board
- Typing on a keyboard
- Driving a car
- Performing surgery
- Playing a musical instrument

Why does psychomotor learning matter in engineering technology?

In engineering technology, psychomotor learning is crucial for:

- Operating tools and machinery
- Assembling components with precision
- Troubleshooting hardware
- Performing lab experiments and fieldwork

It's the **hands-on mastery** that complements theoretical knowledge.

Class Activity 1

Think of the three categories of learning cognitive learning, psychomotor learning, and affective learning.

1. What are some examples of learning you have experienced in each of the three categories?
2. Certainly, you have acquired new knowledge (cognitive learning) and developed dexterity in some area such as typing or sketching or dancing (psychomotor learning). Do you have more difficulty coming up with examples under the affective learning category?
3. Are you open to learning in areas such as enjoying, respecting, and supporting others?

Class Activity 2

Review the following 9 attributes presented below that the ABET Engineering Criteria mandates engineering graduates.

1. Which primarily involves cognitive learning?
2. Which is most likely to involve psychomotor learning?
3. What psychomotor skills might you acquire?
4. Which of the attributes are most likely to involve affective learning?
5. Which affective skills might you acquire?

3.5. ABET Attributes of Engineering Graduates

The ABET attributes in Engineering and Technology graduates are:

a. An ability to apply knowledge of mathematics, science, and engineering
b. An ability to design and conduct experiments, as well as to analyze and interpret data
c. An ability to design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability
d. An ability to function on multi-disciplinary teams
e. An ability to identify, formulate, and solve engineering problems
f. An understanding of professional and ethical responsibility
g. An ability to communicate effectively

h. The broad education necessary to understand the impact of engineering solutions in a global, economic, environmental, and societal context
i. A recognition of the need for, and an ability to engage in, lifelong learning

3.6. How do students learn?

Students learn through a blend of cognitive, emotional, and experiential processes that help them absorb, retain, and apply knowledge. It's not just about listening or reading; it's about **actively engaging** with ideas and making meaningful connections. The process of learning new knowledge and intellectual skills (cognitive learning) involves two steps: 1) receiving new knowledge, and 2) processing that new knowledge.

There are two key pieces to the way you receive new knowledge: The type of info you prefer and the sensory channel through which you most effectively observe external information.

There are two types of learners:
Sensing vs. Intuitive learners represent two distinct ways students prefer to take in and process information, each with their own strengths and challenges. Understanding the difference can help you tailor your learning strategies, especially in engineering technology where both detail and big-picture thinking matter.

Key Differences
When it comes to learning styles, individuals often process information in distinct ways based on their cognitive preferences. Two common types are sensing and intuitive learners, each with unique characteristics that influence how they absorb and apply knowledge. Sensing learners tend to focus on concrete facts, details, and step-by-step procedures, excelling in memorization, precision, and hands-on tasks. In contrast, intuitive learners are drawn to abstract ideas, patterns, and theoretical concepts, thriving in innovation and big-picture thinking. While sensing learners prefer real-world applications and may struggle with abstract material, intuitive learners favor future-oriented ideas but can sometimes overlook important details. Understanding these key differences can help educators tailor instruction to better meet the needs of diverse learners.

- Focus:

    - Sensing learners: concrete facts, details, procedures.
    - intuitive learners: Abstract ideas, patterns, possibilities.
- Strengths:
    - Sensing learners: Memorization, hands-on work, precision.
    - intuitive learners: Innovation, conceptual thinking, theory.
- Preferred content:
    - Sensing learners: Real-world applications, step-by-step methods.
    - intuitive learners: Big-picture concepts, future-oriented ideas.
- Challenges:
    - Sensing learners: May resist abstract or theoretical material.
    - intuitive learners: May overlook details or make careless errors.

When it comes to learning, processing information involves more than simply absorbing facts—it also includes how individuals prefer to handle that information and how they move toward deeper understanding. These two aspects—personal processing preferences and progression toward comprehension—play a critical role in shaping how effectively someone learns. Research has shown that learners vary widely in their approaches to both, highlighting the importance of recognizing and adapting to these differences in educational settings. Understanding these variations can lead to more personalized and effective learning experiences.

There are two aspects to processing that knowledge:

- The way you prefer to process information and the way you progress toward understanding.
- Study has shown that learners differ with regard to their favorites in these two areas.

Active learners. Active learners tend to process information while doing something active with it. Consequently, active learners think out loud, try things out, and prefer group work.
Sitting through lectures is particularly hard for active learners.

Reflective learners. Reflective learners prefer to think about information quietly first. They want to understand or think things through before attempting to do anything themselves. They tend to prefer working alone.

Core Principles of How Students Learn

- Active Engagement: Students learn best when they participate—discussing, solving problems, and experimenting.

- Connection to Prior Knowledge: New information sticks better when linked to what students already know.
- Cognitive Load Management: Learning is most effective when information is delivered in manageable chunks.
- Practice & Feedback: Repetition and timely feedback help reinforce learning and correct misunderstandings.
- Motivation & Emotion: Positive emotions and a sense of purpose boost attention and retention.

In Engineering Technology

For someone like you, learning often means:

- **Hands-on experimentation** to connect theory with practice
- **Collaborative problem-solving** to mirror real-world engineering teams
- **Reflective thinking** to understand the impact of technology on society

Many students enter engineering programs with high hopes but unknowingly set themselves up for academic struggles by making common mistakes early on. Some of the early mistakes made by freshman students include:

- Assume engineering study will be like high school
    - Strategies for overcoming them: Work to understand and adjust to the differences between high school and college-level engineering study.
- Program yourself for failure through too many commitments.
    - Strategies for overcoming them: Create a life situation that enables you to devote adequate time and energy to your studies.
- Spend little time on campus
    - Strategies for overcoming them: Immerse yourself in the academic environment of the institution.
- Neglect studying.
    - Strategies for overcoming them: Schedule study time. Devote significant time and energy to studying.
- Delay studying until test is announced.
    - Strategies for overcoming them: Master the material presented in each class prior to next class.
- Study 100% alone.
    - Strategies for overcoming them: Study collaboratively with other.
- Come to each lecture unprepared.
    - Strategies for overcoming them: Review notes, read text, and attempt problems prior to each lecture.

- Avoid professors
    - Strategies for overcoming them: Interact regularly with professors outside the classroom.
- Cut classes and/or don't get the most out of lectures.
    - Strategies for overcoming them: Attend classes and practice good listening skills. Ask questions in class.
- Fail to take notes or take notes but fail to use the notes properly in the learning process.
    - Strategies for overcoming them: Take effective notes and use a systematic learning methodology to study from notes.
- Skim over the material in an assigned chapter in a rush to get to the assigned homework problems.
    - Strategies for overcoming them: Use reading for comprehension methodology (see Section 5.1) to understand the general concepts thoroughly before attempting problems.
- Fail to solve assigned problems. Don't approach problems using a systematic problem-solving method.
    - Strategies for overcoming them: Solve not only assigned problems but extra problems; use systematic problem solving methods.

Class Activity 3

1. When you receive new knowledge, what type of information do you prefer?
2. Would you describe yourself as a sensing learner or an intuitive learner?
3. Do you prefer to receive information visually or verbally?
4. How do you prefer to process new information — by actively doing something with it, or by reflecting it first?
5. Do you prefer to progress toward understanding through small, logical steps or through larger jumps in information?
6. Which of the common mistakes made by freshman students have you made yourself or believe you might make? What steps will you take to avoid making them in the future?

References

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